

Newly-Constructed Bidirectional DC/DC Converter Topology with High Voltage Conversion Ratio for Vehicle to DC-Microgrid (V2DCG) System

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Abstract—The objective of this paper is to present a novel bidirectional dc/dc converter topology with a high voltage conversion ratio for electric vehicle (EV) batteries connected to dc-microgrid system. In this study, an unregulated level boost converter (ULC) cascaded with a two-phase interleaved buck-boost charge-pump converter is introduced to achieve a high conversion ratio with very simple control circuits. In discharge mode, the topology acts as a two-stage voltage-doubler boost converter to achieve high step-up conversion ratio (48V to 400V). In charge mode, the converter acts as two cascaded voltage-divider buck converters to achieve high voltage step-down conversion ratio (400V to 48V). The steady-state analysis and numerical simulation are made to verify the performance of the proposed newly constructed bidirectional converter.

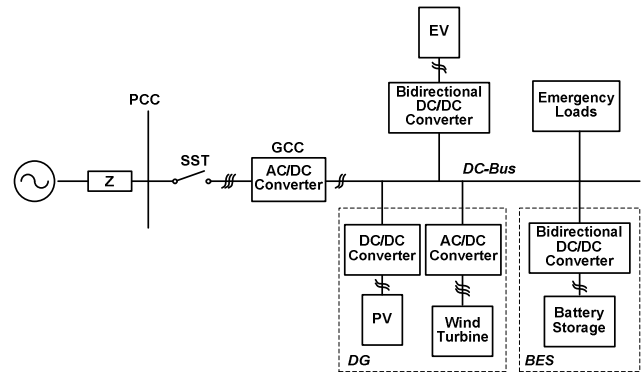


Figure 1. A structure of dc-microgrid.

I. INTRODUCTION

In recent years, due to the shortage of fossil fuels and serious environment problems, much effort has been focused on the development of environmentally friendly distributed generation (DG) technologies [1]-[4]. It is well known that when many DGs are connected to utility grids, they can cause problems such as voltage rise and protection problem in the utility grid [5]-[8]. To solve these problems, new concepts of electric power systems are proposed, and dc-microgrid is one of the solutions [9]-[12]. Fig. 1 shows the structure of dc-microgrid, it includes grid-connected converter (GCC), wind-turbine based generation system, PV generation system, battery energy system (BES), electric vehicle (EV) and local emergency loads. The function of GCC is to keep dc-bus voltage constant. In order to ensure the reliability of operation for dc-microgrid, a mass of battery energy storage system usually be accessed into the system. The electric vehicle can provide ancillary services for dc-microgrid, called vehicle to dc-microgrid, which makes clean and efficient electric-powered transportation possible by allowing electric vehicles to power and be powered by the grid. In such applications, the bidirectional dc/dc converter plays an important role in system. Briefly speaking, the voltage difference between the electric vehicle battery/BES and the dc-bus is large; thus, a bidirectional dc/dc converter with a high step-up/down voltage conversion ratio is required.

The conventional buck-boost bidirectional converter has simple circuit structure; however, the voltage conversion ratio is significantly limited because of the effect of parasitic elements [13], [14]. In previous studies, the bidirectional converter has been classified into two categories, namely, isolated [15]-[18] and non-isolated [19]-[23]. Moreover, the high voltage conversion ratio of bidirectional dc/dc converters is achieved by adjusting the turn ratios of the isolated transformer, such as flyback-type, forward-flyback-type. These converters have a simple and low-cost configuration [15], [18]. However, it is only suitable in low-power applications because of its high voltage stress and leakage inductance energy. Additionally, using high-voltage transformers with a large turn-ratio also introduces several problems. For example, the leakage inductance and the parasitic capacitance formed by the secondary winding of the transformer may cause voltage and current spikes and increases loss and noise that will degrade the system performance significantly and damage circuit components. Prior arts of a non-isolated bidirectional converter, the three-level bidirectional converter has only half of the voltage stress on the switch compared with conventional bidirectional converters. However, due to the voltage conversion ratio is the same with that of the conventional buck-boost bidirectional converter; thus, the three-level bidirectional converter is not suitable for the large voltage difference between the dc-bus

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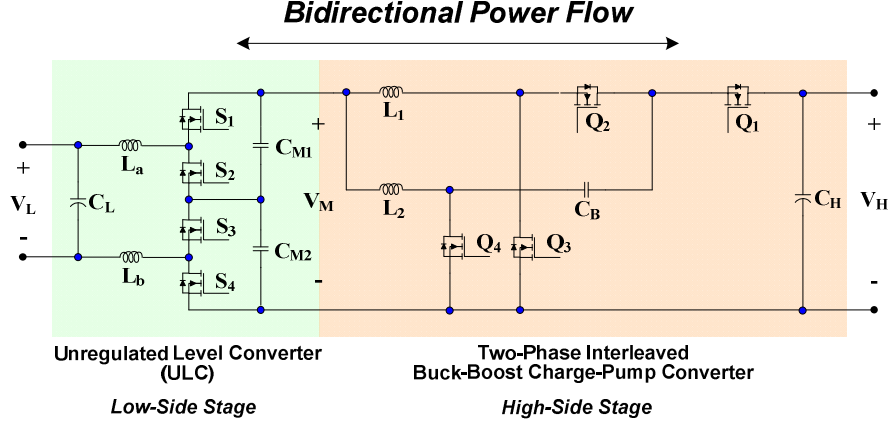


Figure 1. Proposed newly-constructed bidirectional dc/dc converter.

and the battery applications [19]. Compared with isolation transformer based converters, bidirectional converters with coupled-inductors for non-isolation applications possess simpler winding structure and lower conduction loss [20]-[22]. However, the energy stored in the leakage inductor of the coupled inductor causes a high voltage spike on the power switches. Furthermore, although the extreme duty cycle is avoided, the input current ripple is large due to their single-phase operation which renders these converters unsuitable for high power and high current applications.

In this paper, a multi-phase interleaved, bidirectional converter with low switch voltage stress and high voltage conversion ratio for vehicle to dc-microgrid system applications is presented. An ULC cascaded with a two-phase interleaved buck-boost charge-pump converter is introduced to achieve a high conversion ratio with very simple control circuits. In discharge mode, the topology acts as a two-stage voltage-doubler boost converter to achieve high step-up conversion. In charge mode, the converter acts as two cascaded voltage-divider buck converters to achieve high voltage step-down conversion. Due to the lower duty ratio, not only the output voltage regulation range can be further extended but also the resulting conduction loss can be further reduced due to the shorter conduction period. In addition, the proposed two-stage structure benefits reducing voltage stress of active switches, which enables one to adopt lower voltage rating devices to further reduce both switching and conduction losses. Accordingly, because of the device voltage stress reduction, high efficiency can be achieved.

II. DESCRIPTIONS AND OPERATION PRINCIPLES

Fig. 2 shows the circuit configuration of the proposed bidirectional converter. For achieving the high power density and efficiency, an ULC converter is developed. Because only a small sized high frequency line filter (L_a and L_b) is required, it can substantially boost the power density of the low-side stage.

Furthermore, by leaving the voltage regulation to another high-side stage, proposed bidirectional converter for the low-side stage with fixed 2:1 under charge mode operation or 1:2

conversion ratio under discharge mode operation, can achieve high efficiency with a relatively low input voltage in whole load range. For the high-side stage, the structure of two-phase interleaved buck-boost charge-pump converter is similar to a conventional buck-boost converter except two active switches in series and a charge-pump capacitor employed in the power path. The circuit structure is simple and it can reach the same high voltage ratio with lower duty cycle. Therefore, it can reduce the conduction loss of the switch, to further upgrade the efficiency of the whole bidirectional converter.

The proposed converter can transmit energy in both directions. When the energy flows from V_H to V_L , it operates in charge mode (i.e. buck operation); Q_1 and Q_2 are controlled to regulate the output. Thus, Q_1 and Q_2 are defined as the active switches, while Q_3 and Q_4 are the passive switches. The passive switches work as synchronous rectification. When the energy flows from V_L to V_H , it operates in discharge mode (i.e. boost operation); Q_3 and Q_4 are controlled to regulate the output. Thus, Q_3 and Q_4 are defined as the active switches, while Q_1 and Q_2 are the passive switches.

Before the analysis, the following assumptions should be made as follows: 1) the proposed converter is operated in continuous-conduction-mode (CCM); 2) capacitors C_H and C_L is large enough to be considered as a voltage source; 3) the middle-link voltage $V_M = V_{M1} + V_{M2}$ is treated as a pure dc and considered as constant; 4) the two inductor L_1 and L_2 have the same inductor L_s ; 5) all power semiconductors are ideal; 6) the charge-pump voltage V_{CB} is treated as a pure dc and considered as constant.

A. Charge Mode Operation

Fig. 3 and Fig. 4 show the circuit configuration and typical waveforms of the proposed bidirectional converter in charge mode operation, respectively. It can be seen that switches Q_1 and Q_2 are driven with the phase shift angle of 180° ; Q_3 and Q_4 work as synchronous rectification.

In the charge mode, when S_1 , S_3 are turned on and S_2 , S_4 are turned off; or else S_2 , S_4 are turned on and S_1 , S_3 are turned off. The low-side voltage V_L is half the middle-link voltage V_M , i.e. $V_L = 0.5V_M$.

$$L_1 \frac{di_{L1}}{dt} = L_2 \frac{di_{L2}}{dt} = V_M = 2V_L \quad (6)$$

Mode 2 [t_1 - t_2]: For this operation mode, the interval time is $(1-D_b)T_{sw}$. Switch Q_1 , Q_3 remains conducting and Q_2 , Q_4 are turned off. The voltages across inductances L_1 and L_2 can be represented as following equations.

$$L_1 \frac{di_{L1}}{dt} = V_M = 2V_L \quad (7)$$

$$L_2 \frac{di_{L2}}{dt} = V_M - V_H + V_{CB} = 2V_L - V_H + V_{CB} \quad (8)$$

Modes 3 [t_2 - t_3]: In this situation, both Q_3 and Q_4 are turned on; the circuit operation is same as mode 1.

Mode 4 [t_3 - t_4]: During this operation mode, the interval time is $(1-D_b)T_{sw}$. For the low-side stage, switches Q_1 , Q_3 are turned off and Q_2 , Q_4 are turned on. The energy stored in inductor L_1 is now released energy to charge-pump capacitor C_B for compensating the lost charges in previous modes. Also, the output power is supplied from the capacitor C_H . The voltages across inductances L_1 and L_2 can be represented as follows.

$$L_1 \frac{di_{L1}}{dt} = V_M - V_{CB} \quad (9)$$

$$L_2 \frac{di_{L2}}{dt} = V_M \quad (10)$$

III. STYLING STEADY-STATE ANALYSIS

A. Voltage Conversion Ratio

The energy of proposed bidirectional converter can flow in both directions, so there is no discontinued-current mode in operation stages.

In charge mode, V_H is the input and V_L is the output. According to (1)-(5) and based on the voltage-second balance principle in L_1 and L_2 , the voltage conversion ratio M_d in the charge mode can be derived as

$$M_d = \frac{V_L}{V_H} = \frac{D_d}{4} \quad (11)$$

where D_d is the duty cycle of the active switches Q_1 and Q_2 . As can be seen, the voltage conversion ratio in charge mode is one-fourth of that of the conventional buck converter.

Similarly, in discharge mode, V_L is the input and V_H is the output. According to (6)-(10) and based on the voltage-second balance principle in L_1 and L_2 , the voltage conversion ratio M_b in the discharge mode can be derived as

$$M_b = \frac{V_H}{V_L} = \frac{4}{1-D_b} \quad (12)$$

where D_b is the duty cycle of the active switches Q_3 and Q_4 . As can be seen, the voltage conversion ratio in discharge mode is four times of that of the conventional boost converters.

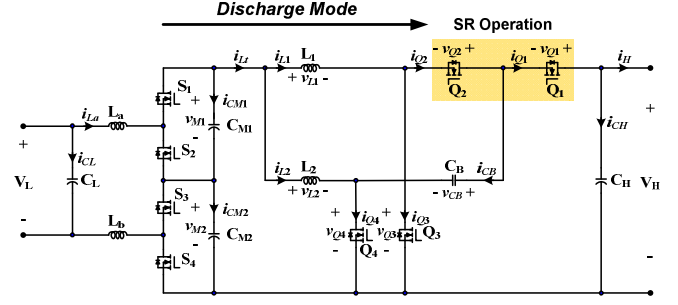


Figure 5. Circuit configuration of the proposed converter in the discharge mode operation.

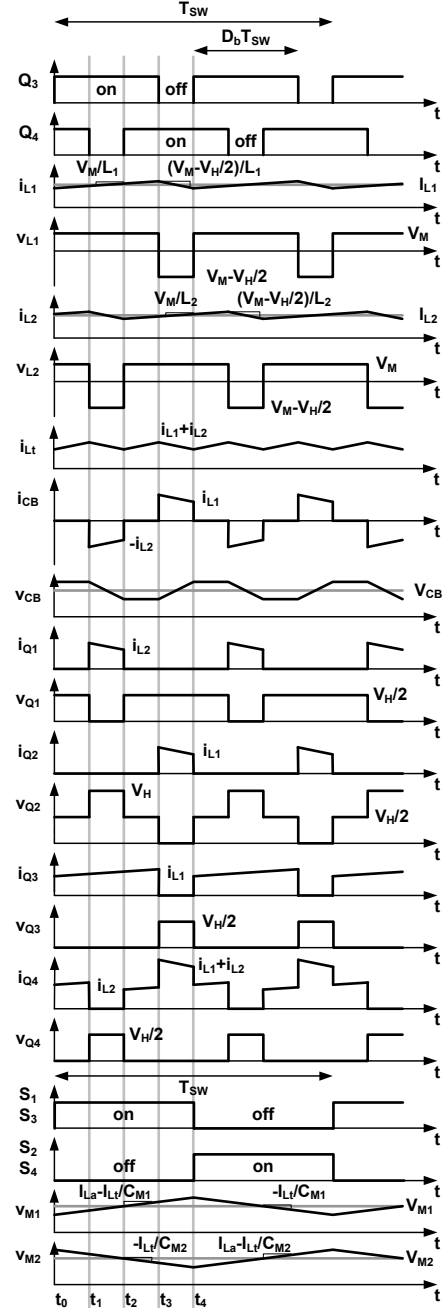


Figure 6. Typical waveforms of the proposed topology in the discharge mode operation.

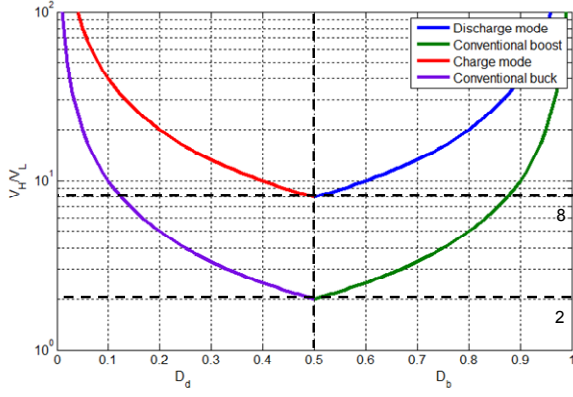


Figure 7. Voltage conversion ratio characteristic of the proposed converter.

Fig. 7 shows the ideal voltage conversion ratio characteristic as a function of duty cycle. It is seen from Fig. 7 that, much widely voltage conversion ratio can be achieved than that of the conventional buck-boost bidirectional converter [14].

B. Voltage Stress of the Switches

Whatever the ULC works as back or front-end stages, the open circuit voltage stress on the switches $S_1 \sim S_4$ of ULC is equal to low-side input voltage V_L , as follows

$$V_{S1,\max} = V_{S2,\max} = V_{S3,\max} = V_{S4,\max} = V_L. \quad (13)$$

The particular inherent feature of the ULC benefits the low conduction losses can be achieved by adopting the low-voltage MOSFETs. For the high-side stage, based on the aforementioned operation analyzes, the open circuit voltage stress of switches $Q_1 \sim Q_4$ can be obtained directly as follows

$$V_{Q1,\max} = V_{Q3,\max} = V_{Q4,\max} = \frac{V_H}{2} \quad (14)$$

$$V_{Q2,\max} = V_H \quad (15)$$

C. Inductor Current Ripple

The proposed converter can operate not only in charge mode but also in discharge mode. Thus, the inductor can be calculated in either charge or discharge mode.

According to (1)-(5), the total ripple current of the inductor of the proposed converter in charge mode can be expressed as

$$\Delta i_{L_t}|_{\text{charge}} = \frac{V_H T_{sw}}{L_s} (0.5 - D_d) D_d \quad (16)$$

Similarly, in discharge mode, according to (6)-(10), the total ripple current of the inductor of the proposed converter in discharge mode can be expressed as

$$\Delta i_{L_t}|_{\text{discharge}} = \frac{V_H T_{sw}}{L_s} (D_b - 0.5)(1 - D_b) \quad (17)$$

Fig. 8 shows the normalized ripple current of the inductor of the proposed converter and the conventional buck-boost bidirectional converter, where the inductor and the switching frequency of the two converters are equal, respectively.

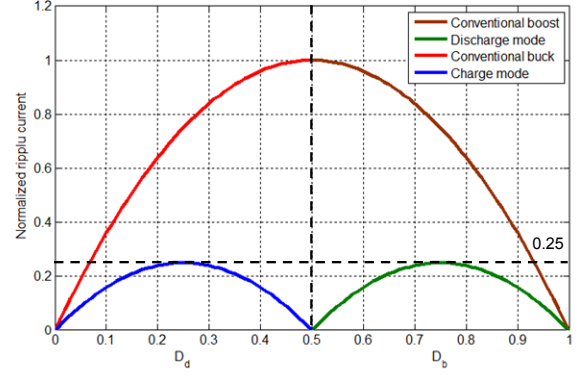


Figure 8. Comparison of the normalized ripple current of the inductor between the proposed converter and the conventional buck-boost bidirectional converter.

The ripple current of the conventional buck-boost bidirectional converter under $D = 0.5$ is normalized as one. It can be seen that from Fig. 7, the maximum ripple current of the inductor of proposed converter is only one-fourth of that of conventional buck-boost bidirectional converter. On the other and, if the ripple currents are equal, the inductor of the proposed converter is only one-fourth of that of conventional buck-boost bidirectional converter, which means that the proposed converter BDC has a better dynamic response.

D. Boundary Conduction Mode

The boundary normalized inductor time constant $\tau_{L,B}$ can be defined as follows

$$\tau_{L,B} = \frac{L f_{sw}}{R} \quad (18)$$

During boundary conduction condition (BCM) operation, the input current of the proposed converter can be derived as

$$I_L = \frac{4V_L}{L_s f_{sw}} (1 - D_d) \quad (19)$$

Substituting (19) into (18), the boundary normalized time constant in charge mode can be expressed as follows

$$\tau_{Ld,B} = 4(1 - D_d) \quad (20)$$

where R_L is low-side input equivalent resistance. Similarly, in discharge mode, the input current of the proposed converter can be obtained as follows

$$I_L = \frac{4V_L}{L_s f_{sw}} D_b \quad (21)$$

The boundary normalized time constant in discharge mode can be expressed as follows

$$\tau_{Lb,B} = 4D_b \quad (22)$$

Fig. 9 shows the plots of boundary normalized inductor time constant curves $\tau_{Ld,B}$ and $\tau_{Lb,B}$ in charge and discharge modes.

The proposed converter in the charge mode operates in CCM when τ_{Ld} is designed to be higher than the boundary curve of $\tau_{Ld,B}$. The proposed converter in the discharge mode operates in DCM when τ_{Lb} is selected to be lower than the boundary curve of $\tau_{Lb,B}$.

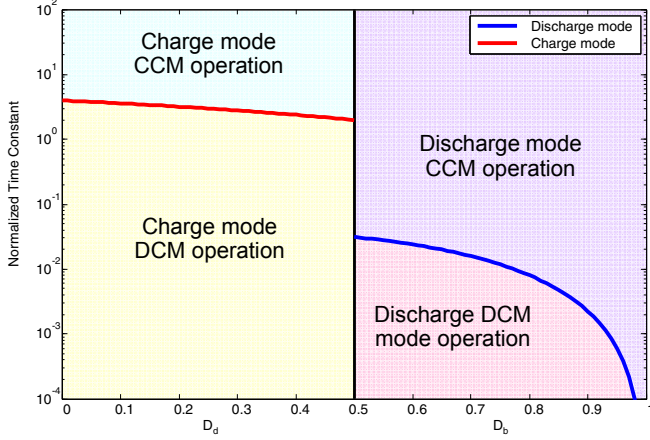


Figure 9. Normalized boundary inductances time constant in charge and discharge modes.

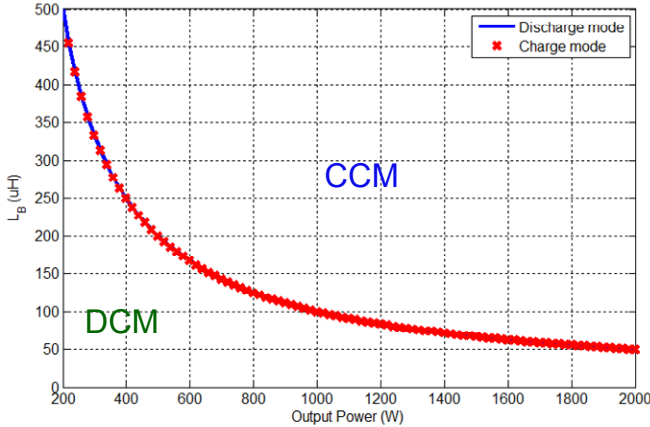


Figure 10. Boundary inductances in various power conditions.

The proposed converter can operate not only in charge mode but also in discharge mode, the boundary inductance can be derived from (19) and (21) as follows:

$$L_{d,B} = \frac{4(1-D_d) V_L^2}{f_{sw} P_{out}} \quad (23)$$

$$L_{b,B} = \frac{4D_b V_L^2}{f_{sw} P_{out}} \quad (24)$$

where P_{out} is output power.

Fig. 10 shows the boundary inductances curve of the proposed converter in charge and discharge modes. If the inductance is selected to be larger than the boundary inductance, the proposed converter will operate in CCM.

IV. NUMERICAL SIMULATION RESULTS

In order to illustrate the performance of the proposed converter, a prototype circuit is simulated. Parameters and specifications of the proposed converter are given as below:

- 1) High-side voltage V_H : 400V;
- 2) Low-side voltage V_L : 48V;
- 3) Rated power: 1kW;
- 4) Switching frequency 50kHz;
- 5) Capacitors: $C_H=C_L=33\mu\text{F}$, $C_{M1}=C_{M2}=33\mu\text{F}$, $C_B=10\mu\text{F}$;
- 6) Inductors: $L_1=L_2=L_s=300\mu\text{H}$; $L_a=L_b=1.5\mu\text{H}$.

Simulated waveforms in the charge and discharge modes are shown in Fig. 11 and Fig. 12, respectively. It can see that the voltage stresses on the switches $S_1 \sim S_4$ are equal to low-side voltage V_L . Also, the waveforms of all the switch voltage stress agree with the analysis of operating principle.

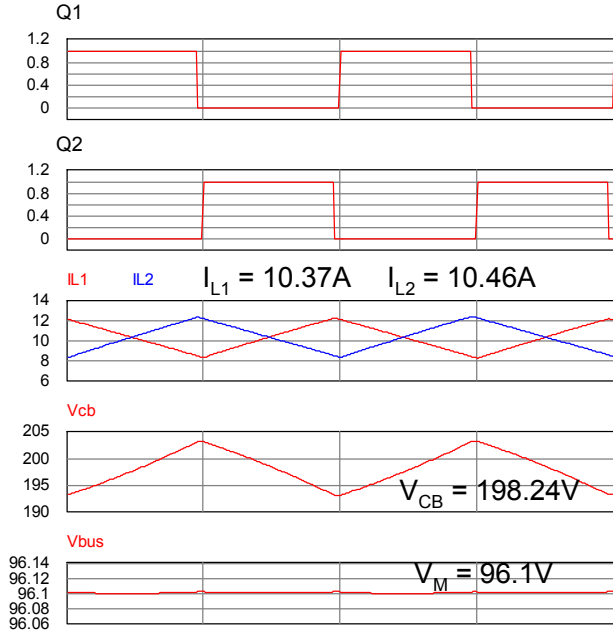
Fig. 11b shows the waveforms of the low-voltage side V_L and filter inductor current i_{La} . It can be seen that the low-voltage side V_L is well regulated at 48V and the proposed converter is operated in CCM in the charge mode. The waveforms in the discharge mode of the high-side voltage V_H , and inductor currents $i_{L1} \sim i_{L2}$ are shown in Fig. 12b. One can see that the high-side voltage V_H is well regulated at 400V and $V_{Q3}=V_{Q4}=V_H/2$.

V. CONCLUSIONS

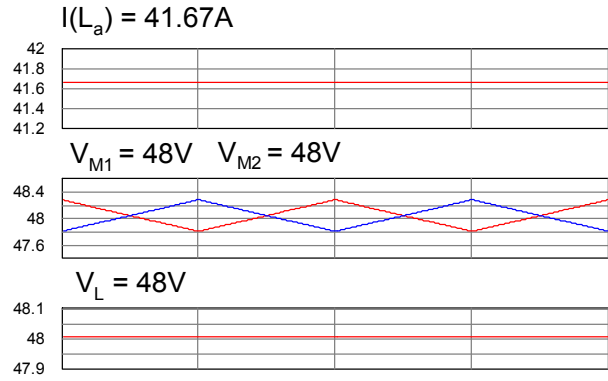
A novel bidirectional dc/dc converter topology with a high voltage conversion ratio is proposed. In this study, an ULC and a two-phase interleaved buck-boost charge-pump converter is introduced to achieve a high conversion ratio. Thanks to the ULC, high power density and efficiency in all load range make the proposed bidirectional converter promising to the two-stage power architecture. Furthermore, the two-phase interleaved buck-boost charge-pump circuit is adapted to achieve much higher voltage conversion ratio and avoid operating at extreme duty ratio. Additional charge-pump capacitor is added for the two-phases for reducing the voltage stress of active switches which enables one to adopt lower voltage rating devices to further reduce both switching and conduction losses.

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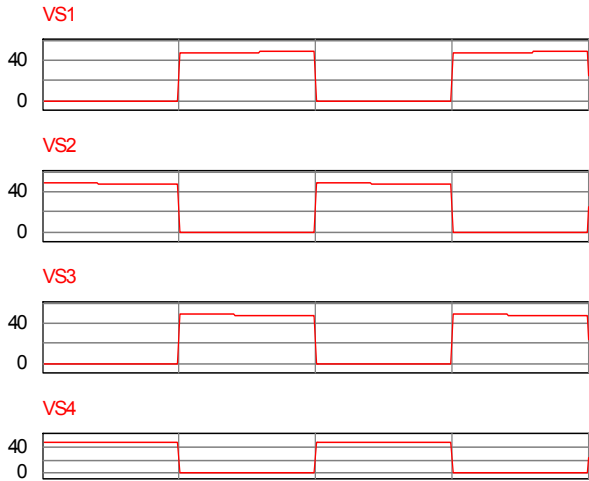
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(a)

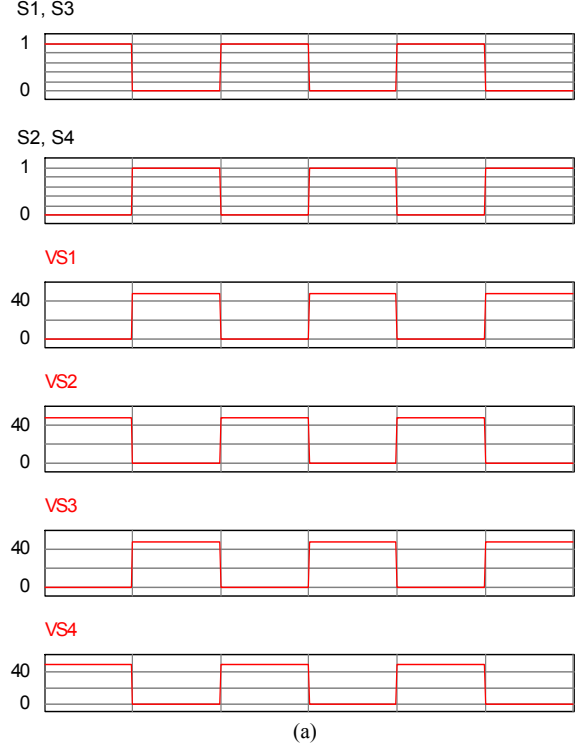


(b)

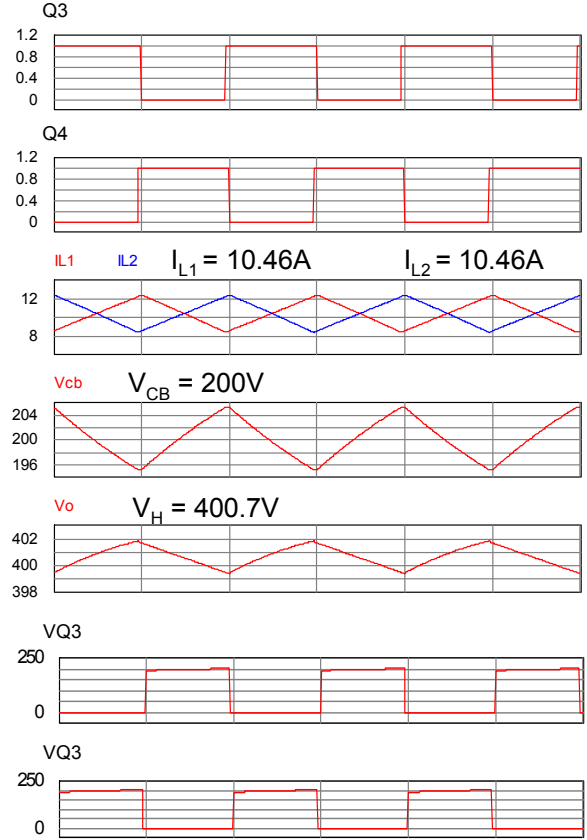


(c)

Figure 11. Waveforms of the proposed converter in the charge mode at full load: (a) gate signals of $Q_1 \sim Q_2$, two-phase inductor currents i_{L1} , i_{L2} , charge-pump voltage V_{CB} and middle-link voltage V_M ; (b) low-side current i_{La} , middle-link capacitor voltages $V_{M1} \sim V_{M2}$, and low-side voltage V_L ; (c) switches voltages of S_1 , S_2 , S_3 , S_4 ;



(a)



(b)

Figure 12. Waveforms of the proposed converter in the discharge mode at full load: (a) gate signals of S_1 , S_2 , S_3 , S_4 and switches voltages of S_1 , S_2 , S_3 , S_4 ; (b) gate signals of $Q_1 \sim Q_2$, two-phase inductor currents i_{L1} , i_{L2} , charge-pump voltage V_{CB} , high-side voltage V_H , and switches voltages of Q_3 , Q_4 .

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